

# DA 221: Introduction to Artificial Intelligence

January - May 2024

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## Lab: Assignment 1

Due: Jan 25, 2024

(To be Completed in 02 Lab Classes)

Max Marks: 25

### Experimenting with Heuristic Search using 8-puzzle

1. As discussed in the class, this **assignment is based on experimenting with some heuristic search algorithms**. We shall use the 8-puzzle game. For the experimentation, you would need to create 10 random 8-puzzle instances (i.e., define start and *reachable* goal states). Each of these are to be solved using variants of A\* algorithm.
2. You may write your own code or tailor code available from other sources for the 8-Puzzle; the A\* algorithm is to be used with different heuristics and need to have possibility of recording the experimental parameters of interest as detailed below.
3. You must cite any kind of help from elsewhere. Each algorithm should be run on the same set of problems for the comparison to be fair.

#### **Q1. Find if the 8-puzzle instance is solvable?**

Note that the 8-puzzle states are divided into two disjoint sets, such that any state is reachable from any other state in the same set, which no state is reachable from any state in the other set. Before you proceed with solving the 8-puzzle instance using the A\* algorithm, it is mandatory to you check (either manually or as part of your 8-puzzle solver implementation) that the instance is solvable.

Definition: For any other configuration besides the goal, whenever a tile with a greater number on it precedes a tile with a smaller number, the two tiles are said to be inverted.

Proposition: For a given puzzle configuration, let N denote the sum of the total number of inversions. Then  $(N \bmod 2)$  is invariant under any legal move. In other words, after a legal move an odd N remains odd whereas an even N remains even.

#### Example

Start State

|   |   |   |
|---|---|---|
| 7 | 2 | 4 |
| 5 | 0 | 6 |
| 8 | 3 | 1 |

Goal State

|   |   |   |
|---|---|---|
|   | 1 | 2 |
| 3 | 4 | 5 |
| 6 | 7 | 8 |

Linear Representation 7; 2; 4; 5; 6; 8; 3; 1 - Ignore the 'blank'

|            |   |              |
|------------|---|--------------|
| Inversions | 7 | 6 Inversions |
|            | 2 | 1 Inversion  |
|            | 4 | 2 Inversions |
|            | 5 | 2 Inversions |
|            | 6 | 2 Inversions |
|            | 8 | 2 Inversions |
|            | 3 | 1 Inversion  |
|            | 1 | 0 Inversions |

Total Inversions      16 – Even

The goal state described above, with no inversions, has  $N = 0$ , and can only be reached from starting states with even N, not from starting states with odd N. Puzzle is solvable.

## Q2. Explore which heuristics work best for 8-puzzle.

Having identified solvable instances of the 8-puzzle, the A\* algorithm is expected to record some vital parameters for each of the solved problem. You are to *at least* record a. the total running time in seconds b. the length of the solution and c. the total number of nodes that are *removed* from the frontier.

The variants of A\* algorithms you should test over *at least* 10 solvable instances are:

- a. A\* using the misplaced tile heuristic.
- b. A\* using the Manhattan distance heuristic.
- c. Any other heuristic of your choice – either *novel* or not discussed in the class!

### Submission Guidelines

- You are expected to present the **SOURCE-CODE together with a README** file to compile and run your code during the scheduled Lab Classes.
- You need to submit **PROGRAM OUTPUT for two SAMPLE SOLUTIONS** with depth of at least 10. For this your program should output each step from the initial to the final state. This is NOT part of the REPORT below.
- A **TWO A4-Page REPORT** detailing discussion on **EVALUATION OF THE HEURISTICS** is to be submitted. The analysis together with justification is to be presented for figuring out the better heuristic for the 8-Puzzle.